Adaptive Entanglement Generation for Quantum Routing

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***Abstract*—Entanglement generation in long-distance quantum networks is a challenging process due to limited resources and the probabilistic nature of entanglement swapping. To maximize the rate of successful requests, existing quantum routing algorithms often rely on computationally expensive methods such as Integer Linear Programming (ILP) to determine which links to entangle and use for end-to-end entanglement generation. However, these approaches fail to meet the latency requirements of real-world quantum networks. In this paper, we propose a reinforcement learning (RL)–based model that rapidly determines which links to entangle in each time slot, replacing the slow ILP-based link-selection phase used in prior algorithms. The proposed Deep Q-learning model is up to $19.2x$ faster than linear programming while maintaining comparable routing performance. Caching of long-lived entanglements further improves throughput, and proactive swapping provides an additional gain of up to 107%. Overall, our approach achieves success rates comparable to state-of-the-art solutions while reducing execution time by more than an order of magnitude, demonstrating a practical path toward real-time quantum routing.**

***Index Terms*—Quantum Network, Entanglement routing, End-to-end entanglement, Deep reinforcement learning**

I. Introduction

Establishing reliable long-distance communication in quantum networks is a fundamental challenge for the realization of large-scale quantum internet [1]. Unlike classical networks, where data is routed over deterministic and persistent links, quantum communication depends on the successful generation and distribution of entanglement between distant nodes. This process is inherently probabilistic and resource-intensive, involving the creation of entangled qubit pairs over individual links and their extension via entanglement swapping across multi-hop paths. As quantum systems scale, intelligently managing the limited and fragile entanglement resources becomes essential for ensuring high-throughput and reliable communication.

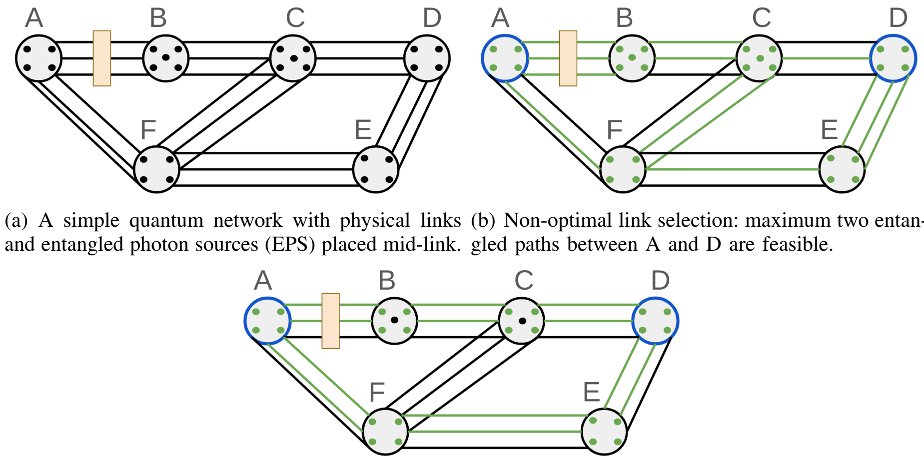
Figure 1(a) shows a simple quantum network that typically consists of quantum repeaters and Entangled Photon Sources (EPS) deployed between nodes. EPS units generate entangled qubit pairs that are transmitted to adjacent nodes through quantum channels, forming the basis for link-level entanglement. Once links are entangled, entanglement swapping operations are performed at intermediate nodes to establish end-to-end entanglement across longer distances. However, the success probability of such operations diminishes with each hop due to the cumulative likelihood of failure at any step in the entanglement generation or swapping process. Therefore, the selection of which links to entangle is a crucial early-stage decision that directly affects the network's ability to successfully serve connection requests.

Prior works on quantum routing [2]–[8] have proposed various optimization-based methods to jointly select links for entanglement and determine paths for connection requests. For example, REPS [4] employs Integer Linear Programming (ILP) for both link selection and path assignment. While such techniques can provide high-quality solutions, they are computationally expensive and do not scale well, taking several minutes even on networks with 30–40 nodes and 10–20 requests. Moreover, while both our approach and prior methods operate in a time-slot-based manner, existing solutions typically discard unused entangled links at the end of each slot, failing to exploit them in subsequent rounds, resulting in inefficient resource utilization and reduced overall throughput.

Existing quantum routing methods tend to rely on slow, rigid link selection techniques that struggle with the probabilistic and dynamic nature of quantum networks. They also typically discard unused entangled links at the end of each time slot, missing the opportunity to reuse valuable resources in subsequent rounds. Without proactive strategies to extend or preserve connectivity over time, these approaches often face scalability challenges and fall short in achieving high routing efficiency (i.e. successful end-to-end entanglement) under realistic conditions.

To address these challenges, we propose Adaptive Entanglement Generator (AEG), a novel framework that introduces reinforcement learning (RL) into quantum routing to tackle the link selection problem more efficiently and intelligently. Our contributions are threefold:

- **Fast Link Selection via Reinforcement Learning:** AEG uses an RL agent to predict which links to entangle, based on real-time network conditions. Unlike ILP-based methods, this approach reduces decision-making delays from several minutes to few seconds without sacrificing performance.
- **Integrating Caching and Proactive Swapping [9]** AEG can optionally reuse long-lived entanglements across time slots and perform proactive swapping to extend connectivity, providing additional performance improvements.



(a) A simple quantum network with physical links and entangled photon sources (EPS) placed mid-link.

(b) Non-optimal link selection: maximum two entangled paths between A and D are feasible.

(c) Optimal link selection: up to four entangled paths between A and D are possible.

Fig. 1. Impact of entanglement link selection on routing performance. The choice of links to entangle directly influences the number of feasible paths, routing reliability, and throughput.

The primary novelty of AEG lies in the RL-based link-selection mechanism, which achieves near-ILP performance at a fraction of the computational cost. Although AEG currently delegates path selection strategy to REPS [4], our experimental evaluation shows that it significantly enhances routing performance. Specifically, AEG achieves $20\times$ faster execution in link selection compared to REPS, while delivering up to 20% higher request satisfaction when combined with entanglement caching. When extended with proactive swapping, the improvement reaches 61% over the baseline REPS algorithm and 107% over random selection of links for entanglement.

In summary, AEG demonstrates that learning-based link selection, combined with temporal reuse and proactive strategies, can substantially improve the scalability and effectiveness of quantum routing, laying the foundation for intelligent and resource-efficient quantum networks.

II. Related Work and Motivation

Numerous routing algorithms have been proposed for quantum networks, including Q-CAST [3], REPS [4], SEER [6], DQRA [10], and SEE [5], with the primary goal of maximizing throughput and reducing idle resources. These approaches typically adopt a two-step solution strategy: first, selecting a set of links for entanglement generation; then, constructing end-to-end paths using the successfully entangled links in a per-time-slot basis. However, they treat each time slot in isolation, without mechanisms for reusing entangled links generated in earlier slots. Additionally, their link selection processes are often either computationally intensive or lack the adaptability needed for large-scale quantum networks.

Several algorithms address link selection for entanglement using heuristic or optimization-based approaches. REPS [4] and SEE [5] use Integer Linear Programming (ILP) for near-optimal link selection: REPS for adjacent nodes and SEE for non-neighbor pairs via all-optical switching. However, ILP becomes computationally impractical as network size and request count grow, limiting real-time applicability. SEER [6] and Q-CAST [3] avoid ILP using greedy shortest-path heuristics, but this rigid path selection reduces adaptability to dynamic conditions. DQRA [10] applies deep reinforcement learning (DRL) for request scheduling but still relies on shortest-path routing, missing the opportunity to exploit DRL for adaptive link selection. Similarly, the RL-based approach in [11] handles only single requests, making it unsuitable for high-load scenarios. These limitations motivate our scalable, DRL-based solution that learns adaptive link selection across multiple requests.

Some studies have begun exploring the temporal persistence of entanglement. Motivated by experimental results showing entanglement can last several seconds [12], [13], the authors of [9] proposed caching unused entangled links and proactively applying entanglement swapping to reduce effective communication distances. They showed performance gains over classical ILP-based algorithms by employing DRL to identify optimal pairs of nodes for proactive swapping. However, their approach is not integrated with scalable link selection methods and does not operate jointly over multiple requests. The overlay approach in [14] builds a static backbone using long-lived entangled links, assuming ideal conditions and low network dynamics. However, it does not address per-request routing or adapt to real-time traffic and resource constraints.

Figure 1 illustrates the critical impact of link selection for entanglement on quantum routing performance. In this network, multiple physical paths exist between source A and destination D, but the effectiveness of routing depends on which links are selected for entanglement generation. As shown in Fig. 1(b), a suboptimal selection allows at most two entangled paths between A and D, constraining throughput and reducing reliability. In contrast, Fig. 1(c) shows that with better link choices, up to four entangled paths can be established, significantly enhancing performance. This example demonstrates that entanglement generation is not an independent preprocessing step but a key decision that directly determines routing success. These observations motivate our proposed learning-based framework, which selects links for entanglement adaptively to maximize path availability and overall network utility.

AEG builds on these findings by replacing the ILP-based link selection of REPS [4] with a reinforcement learning model capable of selecting all potential links for entanglement with a batch prediction technique. This parallel approach allows for scalable and fast decision-making in networks with high traffic and many requests. Moreover, AEG incorporates entanglement caching and proactive swapping strategies to reuse unused entangled links across time slots, addressing a key limitation of earlier methods that discard unutilized resources. Through this combined design, AEG significantly enhances scalability, adaptability, and resource efficiency in quantum routing. Unlike our earlier study on proactive swapping [9], this paper focuses on the RL-based replacement of ILP link selection, addressing the primary scalability bottleneck in existing routing frameworks.

III. Adaptive Entanglement Generation

The existing quantum routing algorithms generally adopt a two-step process when handling connection requests. They are (i) selecting links to generate entanglement and (ii) finding path(s) using successfully entangled links to extend the entanglement from link level to end-to-end [3]. Since entanglement generation and swapping operations are probabilistic, quantum routing algorithms choose multiple links to entangle and swap for a given request since connection attempt on primary path may fail. Then, entangled qubit pairs are created for each link using Entanglement Photon Sources (EPS) [4] whose success rate is contingent upon the inherent probability associated with each link. From the pool of successfully entangled links, an optimal path is selected for entanglement swapping. Similar to the entanglement generation, entanglement swapping operation can also fail due to the possibility of failure in the Bell State Measurement (BSM) process. Ultimately, if at least one of the selected paths is successful in creating entanglement on all links and conducting BSMs, the request is deemed to be successful.

Existing quantum routing algorithms discard unused entanglements between consecutive time slots, thus they perform entanglement generation and swapping operations in each round from scratch. However, previous studies showed that entanglement between two qubits can endure for up to ten seconds [13]. Hence, we propose to cache unused entangled links for a few time slots (e.g., 10 time slots) to better utilize resources and improve the performance of routing algorithms [9]. Moreover, state-of-the-art quantum routing solutions rely on expensive computational solutions such as linear programming for optimal link and path selection, which are not scalable especially for large scale networks due to slow nature of these optimization algorithms. Hence, we introduce reinforcement-learning based link selection model to speed up the process.

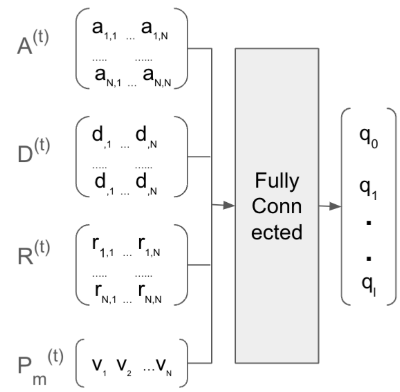


Fig. 2. A sample input and output of the Deep Q-Learning. It takes network topology, distances among nodes, requests and entanglement edges as input and output a list of potential links for entanglement generation

A. Reinforcement Learning-Based Link-Level Entanglement Generation

Limited quantum memories and probabilistic nature of entanglement generation create a significant challenge for establishing end-to-end connections in quantum network. For a given set of requests and resources, Integer Linear Programming (ILP) can be applied to select which link to entangle and swap to meet as many requests as possible. While ILP yields high performance due to transparent mathematical optimization, it can take a long time to find a solution in large scale networks. Hence, it is not suited for problems that require real-time decisions. Reinforcement Learning (RL) based algorithms have been widely used to solve complex optimization problems. A trained RL model can directly predict all potential links quickly by avoiding the cost of repeated calculations. In particular, Deep Q-reinforcement learning (DQRL) is known to attain high performance and scalability with the help of deep learning based predictions. However, RL models require a clear definition of states, actions and rewards function that accurately reflect the core aspects of the networking environment and routing objectives to achieve high performance. In our problem, we define them as follows:

- *States:* The DQRL state space is consist of four components:
  1) *Topology:* The connectivity of the network $G^{(t)}(V, E^{(t)})$ at time slot t is calculated as a subset of the topology G. Only links that not entangled already are considered as edges. Then, we extract the adjacency matrix $A^{(t)} = (a_{i,j})_{1 \leq i,j \leq |V|}$ of $G^{(t)}(V, E^{(t)})$ where $a_{i,j}$ is the number of links between nodes i and j. This is a $N \times N$ matrix where N is the number of nodes.
  2) *Link Distance:* Distance of links impacts successful entanglement generation. Thus, we consider the distances between links in our state and represent them as adjacency matrix $D^{(t)} = (d_{i,j})_{1 \leq i,j \leq |V|}$ where $d_{i,j}$ is the distance between nodes i and j.
  3) *Requests:* The new requests at time slot t along with the remaining requests from previous time slots are

represented a $N \times N$ matrix which is denoted as $R^{(t)} = (r_{s,d})_{1 \le s,d \le |V|}$ where $r_{s,d}$ is the number of requests between source and destination nodes.

4) *Edges:* The m^{th} pair (n_1, n_2) from the set of edges in the topology that will be chosen to create entanglement is represented by a binary vector $P_m^{(t)}$, where each element v_i is either 1 (if i belongs to (n_1, n_2)) or 0 otherwise.

The above four components make up the input state for m^{th} pair $S_m^{(t)} = [A^{(t)}; D^{(t)}; R^{(t)}; P_m^{(t)}]$

- *Output Actions:* The Deep Q neural network outputs a Q-value vector for each link in each time slot $Q_m^{(t)} = [q_0^{(t)}, q_1^{(t)} ....., q_l^{(t)}]$ of length $l+1$ where l is the maximum link capacity between two nodes and the action $a_m^{(t)} = argmax_{i \in 0,1}(q_i^{(t)})$ is the number of links we'll try to generate entanglement for the m^{th} pair or if $a_m^{(t)}$ is greater than maximum number of links between the pair then will try to entangle all the links. Figure 2 shows an example of the input and output.
- *Rewards:* We define rewards and penalty as follows.
  1) We assign a positive reward ($R(s,a) > 0$) for an action (i.e., choosing a link to entangle) if a selected link is used for a request either directly or as part of link segments that is explained in Section III-B.
  2) We assign a negative reward (($R(s,a) < 0$) to an action if selected link expires (i.e., lifespan ends) without being used for a connection request. This penalty ensures that the model will try avoid choosing links that are not likely to be used in the future.

Model Training: Instead of using the traditional train-then-evaluate method, we design the algorithm as an online optimizer with both exploration and exploitation phases. Exploration phase helps the model to stay up-to-date with evolving networking dynamics. The exploitation phase, on the other hand, aims take advantage of current model weights to predicts optimal links to entangle in any specific time slot. In our implementation, we maintain two versions of Q network for model training, a prediction network and a target network. The Q-values are obtained from the prediction network which is trained using the Stochastic Gradient Descent algorithm. The target network's weights are updated periodically with those of the prediction network, typically at every few hundred time steps. The target network is exclusively utilized for forecasting future Q-values. While this process is expected to repeat indefinitely, we used a model with fixed number of iterations to compare results with the state-of-the-arts solutions.

This RL-based link-selection step is the key contribution of AEG, providing a practical, low-latency alternative to ILP without degrading success rate.

B. Proactive Swapping with Cached Entanglements

Existing routing algorithms discard unused entanglements after each time slot, requiring new generation for subsequent rounds. To improve efficiency, AEG reuses these resources through entanglement caching, where unused entangled links are temporarily stored and discarded once their lifetime expires.

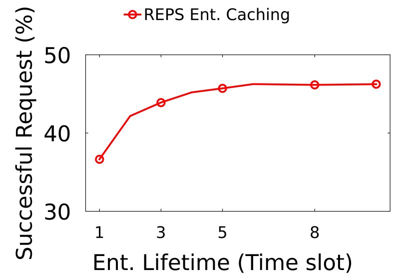


Fig. 3. Impact of entanglement lifetime on the performance of AEG.

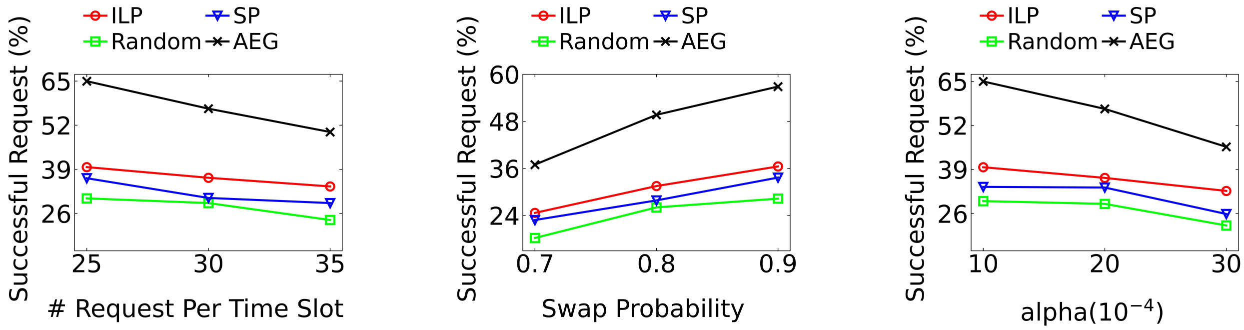
Beyond caching, AEG can optionally incorporate proactive entanglement swapping, a mechanism proposed in our earlier work [9], which combines neighboring cached links into longer segments to reduce path length and improve future routing success. This module operates under resource constraints (e.g., limited repeater capacity) and follows the same Deep Q-Learning training strategy described in Section III-A.

While the caching and swapping modules are inherited from our previous study, their integration here provides complementary temporal optimization to our primary contribution, the RL-based link-selection model,which determines the most effective links for entanglement generation and supplies input to these optional modules.

IV. Evaluations

To evaluate the performance of RL-based link selection and proactive entanglement swapping solutions, we integrated AEG into quantum routing algorithm, REPS [4]. REPS uses two separate ILPs for link selection and path selection operations. We replaced the first ILP in REPS with AEG's RL based link selection algorithm. Then, we incorporated proactive entanglement swapping solution into REPS's path selection algorithm. As a result, proactive entanglement generation receives the output of RL-based link selection algorithm (i.e., the network with a set of successfully entangled links) as input and produces output (a new network with a combination of entangled segments and links). Its output is then consumed by the second ILP of REPS algorithm to choose the optimal paths for given connections.

To ensure fair comparison, we also implemented two additional baseline link selection methods: (i) Random, where links are randomly selected for entanglement generation, and (ii) Shortest Path-Based (SP), where links along the shortest paths between source-destination pairs are prioritized for entanglement generation. For both baselines, we use the same REPS path selection ILP after the entanglement generation phase, allowing all methods to be evaluated under a consistent routing framework. Results show that replacing the ILP link-selection with RL maintains or exceeds REPS throughput while cutting computation time by roughly 20×, confirming the practicality of our approach for online use.



(a) Varying number of requests, AEG outperforms others by up to 107.26%.

(b) Varying swap probability, AEG shows up to 54% gain over ILP.

(c) Varying entanglement success, AEG exceeds baselines by up to 104.63%

Fig. 4. Performance comparison of ILP, random, shortest path and AEG for varying request count, swap probability, and entanglement probability values.

A. Evaluation Methodology

To construct the network, we employed the Waxman model and adhered to the simulation methodology outlined in [3], [4], [6]. In accordance with these simulation parameters, we placed 50 quantum nodes within a rectangular area measuring 2000 km x 4000 km. Each link has a randomly generated qubit transmission capacity between 3-7 and each node has randomly generated quantum memory capacity between 10-14. The probability for entanglement generation was formulated as $P(u,v) = e^{-\alpha \cdot l(u,v)}$, where $l(u,v)$ represents the Euclidean distance between two nodes. A typical value for the parameter α was set to 0.002. With $\alpha = 0.002$ and $l(u,v) = 100km$ the value of $P(u,v) = e^{-\alpha \cdot l(u,v)}$ is 0.819. The success probability for entanglement swapping was configured to 0.9. Our simulations, on average, ran for 200,000 time slots, and the results are averaged over 10 trials for each outcome. In Q-value updating phase of Q-learning, we use learning rate of $\beta = 0.1$ and discount factor of $\gamma = 0.95$.

B. Evaluation Results

1) Entanglement Lifetime: The impact of entanglement caching strategy is dependent on the duration of entanglement. If an entanglement can be sustained for only one time slot, no entangled link can be retained in the cache. However, if entanglement can persist longer, then caching can enhance the performance. We illustrate how entanglement duration impacts the performance of quantum routing in Figure 3. When entanglement is sustained for two time slots, the performance improves by 7%. The improvement rate increases to 19.21% when the entanglement lifetime is set to six or more time slots. It is evident, entanglement caching leads to much better performance, and the success rate plateaus at six time slots, indicating that all cached entangled links are used within six time slots. Given that the entanglement lifetime can last for 10 seconds [13] and most quantum routing algorithms consider each time slot as one (or less) second, we set the default entanglement lifetime as 10 time slots.

2) Request Count: Figure 4(a) illustrates the impact of the number of new requests. As the number of requests increase, we observe a decrease in the performance in terms of serving the requests. AEG outperforms ILP in term of serving requests by 52.55% and random and SP by upto 107.26% and 64.33%. This can be attributed to AEG's entanglement caching and proactive swapping solutions as it helps us to reduce the number of links to entangle and swap at the time of request.

3) Swap Probability: The success of quantum entanglement routing algorithms is significantly influenced by the probability of successfully swapping entangled links. As depicted in Figure 4(b), an increase in the swap probability correlates with improved performance for REPS algorithm. AEG leads up to to 54% improvement over the ILP of the original REPS algorithm. We see our AEG can outperform random and SP by 103.54% and 69.69%. As AEG includes proactive entanglement swapping and entanglement caching on top of RL based link selection for entanglement, this finding emphasizes the impact of our proactive entanglement swapping strategy, particularly in scenarios with higher swap probabilities.

4) Entanglement Generation Probability: The probability of successful entanglement generation affects the performance of quantum routing algorithm. Figure 4(c) shows that as the success rate of entanglement reduces (entanglement success probability has inverse relation to α), the rate of successful requests also decreases. AEG improves the performance over REPS by up to 52.38% and outperforms random and SP by 104.63% and 79.63%. The performance difference is higher when entanglement generation probability is higher since entanglement caching and proactive swapping are able to cache more entangled links and create more entangled segments.

C. Ablation Study

Figure 5 shows the impact of entanglement caching and proactive entanglement swapping for AEG. We observe that RL-based link selection for entanglement generation yields similar performance to ILP of REPS but outperforms random and SP by 37.5% and 17.39%. Entanglement caching with RL based link selection improves the performance of ILP by 18.34%. We get the best performance combining RL based link selection with caching and proactive swapping. While entanglement caching improves the performance by 15-20% over ILP in REPS, Deep Q-learning based proactive entanglement swapping increases the request success rate by up to 52.55% for ILP and by upto 107.26% and 64.33% for

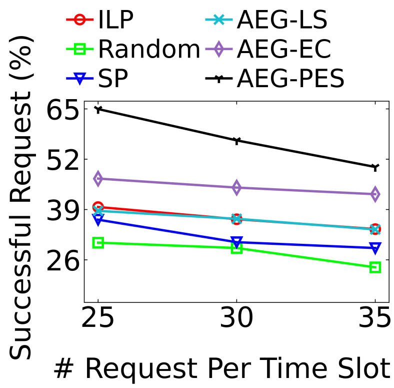


Fig. 5. Impact of link selection, caching, and proactive swapping on AEG performance. AEG-PES yields the highest success rate, significantly outperforming other approaches

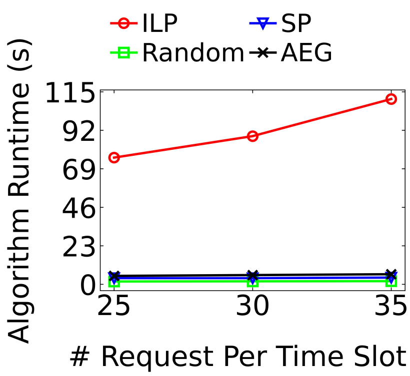


Fig. 6. Execution time for ILP in REPS, random, shortest path and AEG for link selection phase. While REPS relies on Integer Linear Programming, AEG takes advantage of Reinforcement Learning to lower the execution time without sacrificing the performance.

random and SP , demonstrating the effectiveness of proactively swapping entanglement over frequently accessed links.

1) Execution Time: ILP in REPS uses mathematical constraints to obtain near optimal performance [4]. In Figure 4, we show that by replacing ILP with Deep RL, AEG attains a similar performance with significantly lower execution time. The reinforcement learning approach reduces the runtime to select links for entanglement as much as 20 times than the ILP approach in REPS. We observe that with increasing request per time slot, the runtime remains almost constant in our solution compared to linear increase with ILP. While random and SP also executes in linear time, AEG outperforms them in terms of successful requests too.

V. Conclusion

In this paper, we introduced a fast and scalable link selection mechanism for large quantum networks. We applied reinforcement learning (RL) to select potential links for entanglement generation and swapping. RL reduces the execution time by more than 20× compared to integer linear programming based solution. Additionally, by taking advantage of entanglement caching and proactive swapping, we can improve the performance of routing algorithms by around 107%. The evaluation results shows that RL is able to reduce execution time without compromising the performance, which makes it more practical and feasible for real-world quantum networks establishing a critical foundation for deployable large-scale quantum internet architectures.

As a next step, our efforts will concentrate on replacing the remaining complex component, the path selection stage, with an optimized, learning-based approach. We plan to develop an optimal path-finding algorithm that can seamlessly integrate with the RL-based link selection to create a complete, end-to-end quantum routing solution that is both highly performant and computationally lightweight.

Acknowledgement

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